

Case Study

Progression of Joint Destruction In Rheumatoid Arthritis

Group 2

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Declaration

We affirm that the work presented in this report—including the R programs, data analysis, and the accompanying text—is entirely original and has not been copied from any other source. Each member has participated actively and contributed sufficiently in the research and the preparation of this document.

Signed by:

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1 Introduction

Rheumatoid arthritis (RA) is a chronic inflammatory disease that leads to progressive joint damage, significantly impacting patients’ quality of life. The severity of joint destruction is measured in two ways: the Sharp-van der Heijde score (SHS), a quantitative criteria from X-rays on hands and feet of the patients, and a binary classification of disease severity evaluated by the clinicians. This study explores how joint destruction evolves over time in RA patients and examines its association with age, gender, and genetic background while also considering two different periods, using both of the two measurements of outcomes.

This report contains two parts: the main body and the appendix. The main body includes three sections-Descriptive Analysis, Statistical Analysis and Discussion. The Descriptive Analysis section introduces the dataset, discusses the mechanism of missing data, and presents exploratory data analysis (EDA) for both the mean structure and the random effects. The Statistical Analysis section presents the final selected model along with its assumptions and interpretations. The Discussion summarizes the key findings and limitations of this study. The Appendix contains all the analytical methods and procedures, including the statistical software we used, model selection and model diagnostics.

2 Descriptive Analysis

2.1 Statistics Summary

In this study, 500 patients were included between 1990 and 2006, and X-rays were taken at baseline and annually for a 7-year period. The collected dataset contains 2723 records. Table 1 provided an overview of all variables in the dataset. For categorical variables such as Severity, Period, Sex, and SNP, we examined the absolute counts and proportions in each treatment level. For continuous variables such as Sharp index and age, we examined their quantiles.

Table 1: Summary Statistics for All Variables

Variable	Categorical		Continuous		Range
	No.	%	Median	Interquartile Range	
High Severity (1 = yes, 0 = low)	664	24.4	–	–	–
Period A (1 = yes, B = no)	1374	50.5	–	–	–
Male sex (1 = yes, 2 = no)	1245	45.7	–	–	–
SNP0 (0 = yes, 1 = SNP1, 2 = SNP2)	1370	50.3	–	–	–
Sharp index	–	–	9.030	2.679–19.806	0.000–119.452
Visit	–	–	–	–	0–7
Age	–	–	63.6	53.3–77.4	41.2–87.9

2.2 Missing Data

In the dataset, some measurements were not recorded for various reasons including loss of follow-up, relocation, or remission. Missingness is independent of the unobserved outcomes, suggesting that the mechanism is either Missing at Random (MAR) or Missing Completely at Random (MCAR). Therefore, it is recommended to use likelihood-based models to appropriately handle missing data which was taken into consideration during the model-building process.

2.3 Exploratory data analysis

Exploratory data analysis was performed in order to investigate whether mean structure and random effects exist, helping to select the best-fitting model.

2.3.1 Sharp Index

Mean Structure Exploration The comparisons of mean SHS across SNP groups, sex, age categories and visits reveal different SHS progressions for some groups. However, trends for some subgroups are not consistently present (Figures 1 to 2). This suggests that group-level averages may not capture the variability of SHS progression. Also, the pattern of mean SHS progression in time appears non-linear. Some of the groups show an acceleration towards the later visits. These results indicate that quadratic or higher order terms for time may be necessary to capture SHS trajectory over time.

Random Effect Exploration A linear fixed-effects model with non-linear terms is appropriate for reflecting individual SHS changes over time. This approach allows for controlling within subject variability and provides more accurate modeling of SHS progression compared to a model which only uses group level summaries. Strong individual variabilities were observed in Figures 6 and 7.

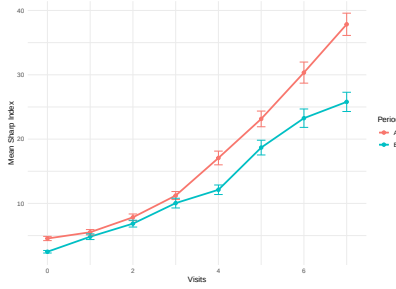


Figure 1: Mean SHS over Visits and Periods

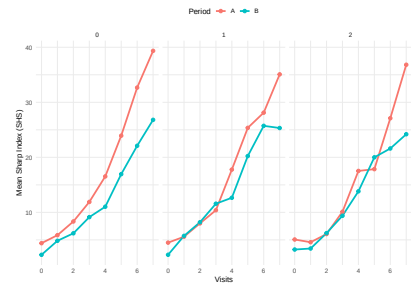


Figure 2: Mean SHS Over Period by SNP

2.3.2 Disease Severity

Mean Structure Exploration As shown in Figures 3, an increasing and non-linear trend is seen over visits for the overall severity. The relationship between severity proportion and covariates was analysed and it reveals a distinct gender difference as females show a higher proportion of severity, which rises sharply starting from the fourth visit (Figures 4), suggesting a potential interaction effect. Additionally, Period A shows a slightly higher proportion than Period B (Figures 5). Other covariates (e.g., age, SNP) do not show a strong pattern.

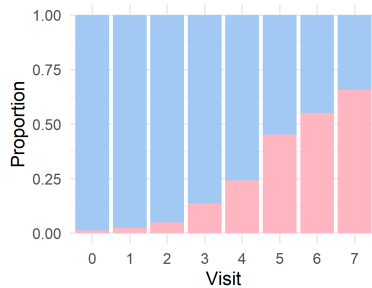


Figure 3: Proportion of severity over visits

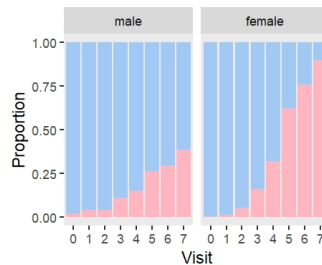


Figure 4: Severity proportions by visit and sex

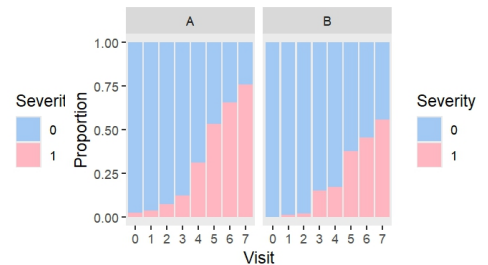


Figure 5: Severity proportions by visit and period

Random Effect Exploration To intuitively explore the necessity of including a random effect, a simple model (with only the main effect of Visit and a random intercept) was initially fitted and used

to predict the probability of high disease severity for each subject. Then, a sample of 100 subjects out of 500 was randomly selected to illustrate within-subject variation. Figures 8 clearly shows the individual differences in predicted severity probabilities over visits. Thus, random effects are expected and included in the model.

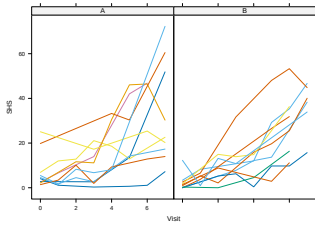


Figure 6: Individual changes in SHS by Period.

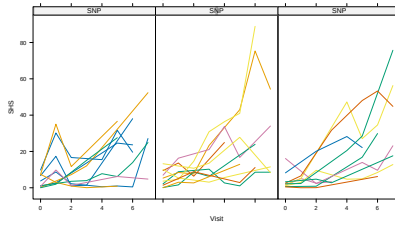


Figure 7: Individual changes in SHS by SNP

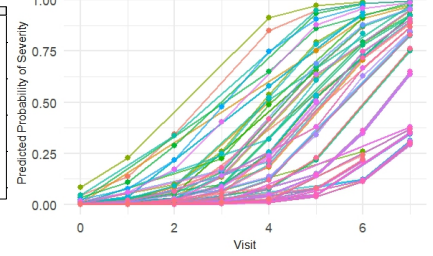


Figure 8: Individual differences in fitted probabilities

3 Statistical Analysis

3.1 Sharp index

Final Model

We followed a three-step procedure to build the model (see Appendix B2 for detailed procedures) and the final model for SHS analysis is a linear mixed effects model, described by the following equation:

$$\log_SHS_{ij} = \beta_0 + \beta_1 Period_i + \beta_2 Sex_i + \beta_3 Visit_{ij} + \beta_4 (Sex_i \times Visit_{ij}) + b_{0i} + b_{1i} Visit_{ij} + \varepsilon_{ij} \quad (1)$$

where, $i \in \{1, \dots, 500\}$: number of subjects, $j \in \{0, \dots, 7\}$: number of visits

$$\begin{bmatrix} b_{i0} \\ b_{i1} \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \sigma_{int}^2 & \sigma_{int,visit} \\ \sigma_{int,visit} & \sigma_{visit}^2 \end{bmatrix}\right), \text{ and } \varepsilon_{ij} \sim N(0, \sigma^2) \text{ are assumed independent of the covariates.}$$

In the fixed-effect part, we included the main effects of period, sex, and visit, along with the interaction between sex and visit (estimated parameters of fixed-effect is shown in Table 2). For random effects, we modeled correlated random intercepts and random slopes (estimated parameters of random-effect is shown in Table 3). The error terms were assumed to be normally distributed, homoscedastic, and independent of the random effects. We verified these assumptions through residual analysis and checked the autocorrelation of the residuals to ensure that no additional correlation structure was needed (see Appendix B3 for details).

Interpretation of Model

In this analysis, Sharp Index (SHS) variable was log transformed as $\log(SHS + 1)$ in order to reduce skewness and stabilize variance. Therefore, all model coefficients are interpreted on the log transformed scale. Estimated coefficients will be interpreted as the expected change in $\log(SHS + 1)$ for a one unit increase in the corresponding predictor, holding all other variables constant.

Visit coefficient is 0.178, indicating that with each additional visit, expected value of $\log(SHS + 1)$ increases by 0.178 units on average. The coefficient for Sex (default category = female) is -0.571, meaning that at baseline (Visit 0), female patients have a 0.571 units lower expected $\log(SHS + 1)$ value than male patients. Similarly, patients in period B have an expected $\log(SHS + 1)$ value that is 0.326 lower than patients in period A, after adjusting for other covariates. The interaction term (Sex x Visit) is 0.248, indicating that the rate of increase in $\log(SHS + 1)$ per visit is 0.248 units higher for females than for males.

Table 2: Fixed Effects Parameter Estimates

	Estimate	Std_Error	t_value	p_value	Lower_CI	Upper_CI
(Intercept)	1.639	0.052	31.554	0	1.537	1.741
PeriodB	-0.326	0.051	-6.412	0	-0.426	-0.226
sexfemale	-0.571	0.062	-9.224	0	-0.693	-0.450
Visit	0.178	0.008	21.023	0	0.161	0.194
sexfemale:Visit	0.248	0.012	21.507	0	0.225	0.270

Table 3: Variance-Covariance for Random Effects with Total Variance

	b_{i0}	b_{i1}	Total
b_{i0}	0.290	-0.012	—
b_{i1}	-0.012	0.004	—
Total	—	—	0.27

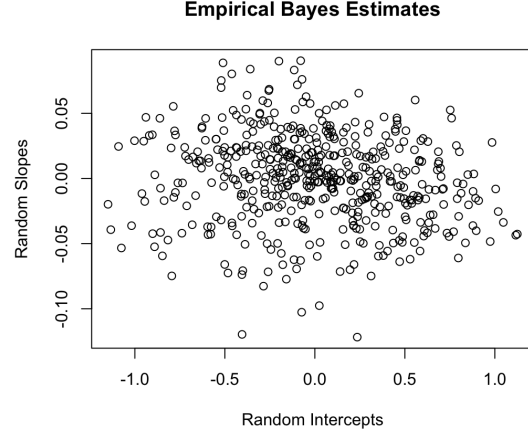


Figure 9: Empirical Bayes Estimates Pattern

The estimated variance-covariance matrix for random effects for SHS models shows variability in the random intercepts. This result indicates that patients differ in their baseline joint destruction levels. In contrast, the variability in the random slopes is minimal which suggests that most patients have similar progression profiles over time. Covariance between the random intercepts and the random slopes is -0.012. Which reflects a weak relationship. This result is confirmed by the empirical Bayes plot (Figure 9). In this plot no clear pattern is observed between subjectwise intercepts and slopes. Overall, model suggests that individual differences in the SHS score are mainly due to baseline severity rather than progression rate in time across patients.

3.2 Disease Severity

To understand how the probability of high disease severity changes over time and how this evolution is influenced by patient characteristics, a generalized linear mixed-effects model (GLMM) was fitted.

The modeling procedure started with an elaborated generalized linear mixed model (GLMM) fitted using the `glmer()` function. The model was then simplified step by step to obtain a parsimonious version under model selection criteria. Models were selected based on LRT and AIC. Variables with non-significant effects based on the Wald test were removed one by one, and the corresponding model was fitted and compared to check its performance. Subsequently, the `mixed_model()` function with Adaptive Gaussian Quadrature was used to obtain accurate estimates of the marginal effects. The detailed modelling and diagnostic procedure is displayed in Appendix C.

Final Model

The final GLMM model includes both fixed effect and random effect:

$$\log \left[\frac{\pi_{ij}}{1 - \pi_{ij}} \right] = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Sex}_i + \beta_3 \text{Period}_i + \beta_4 (\text{Visit}_{ij} \times \text{Sex}_i) + b_{0i} + b_{1i} \text{Visit}_{ij} + \epsilon_{ij} \quad (2)$$

where, $\pi_{ij} = Pr(Severity = 1)$, $i \in \{1, \dots, 500\}$: number of subjects, $j \in \{0, \dots, 7\}$: number of visits.

$$\begin{bmatrix} b_{i0} \\ b_{i1} \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \sigma_{\text{int}}^2 & \sigma_{\text{int,visit}} \\ \sigma_{\text{int,visit}} & \sigma_{\text{visit}}^2 \end{bmatrix}\right), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

In the mean part, the main effects include Visit, Sex, Period, and the interaction of Visit and Sex. For the within subject correlation, both random intercepts and random slopes are included. The error terms are assumed normally distributed, homogeneous and independent of the random effects. The fixed and random effect estimates of the model are summarized in Table 4 and Table 5, respectively.

Table 4: Fixed Effects Parameter Estimates

	Estimate	Std. Error	p-value
Intercept	-6.2552	0.6944	< 2e-16
Visit	0.8516	0.1152	1.45e-13
Sex (female)	-3.0193	0.7450	5.06e-05
Period B	-1.5714	0.3474	6.09e-06
Visit $\times$ Sex (female)	1.3332	0.1948	7.71e-12

Table 5: Random Effects Estimates

	Variance	Std. Dev.	Corr
(Intercept)	12.3201	3.5100	
Visit	0.4009	0.6332	-0.62

Since the primary interest was in the marginal effect of high disease severity in relation to different covariates and their changes over time, the model was refitted using the `mixed_model()` function. The results of the marginal model are summarized in Table 6.

Table 6: Marginal Model Parameter Estimates with Odds Ratios and 95% CIs

	Estimate	Std. Error	p-value	OR	Lower CI	Upper CI
(Intercept)	-3.2706	0.2991	7.93e-28	0.0380	0.0211	0.0683
Visit	0.4881	0.0515	2.68e-21	1.6293	1.4728	1.8024
Sex (female)	-1.9078	0.4329	1.05e-05	0.1484	0.0635	0.3467
Period B	-0.8968	0.2035	1.05e-05	0.4079	0.2737	0.6078
Visit $\times$ Sex (female)	0.6908	0.0835	1.35e-16	1.9953	1.6939	2.3502

Interpretation of Model

Based on the marginal model summary results (Table 6) which provide the population-level estimates, the baseline risk of having high disease severity are low (OR = 0.038, CI: 0.02-0.07); The odds of high disease severity increase over time/Visit (OR = 1.63, CI: 1.47-1.80); The interaction of time/Visit and Sex is significant (OR = 1.99, CI: 1.69-2.35), even though females are low in odds of high severity (OR = 0.15, 0.06-0.35) at baseline, the increase in odds over time is greater for females than males; The odds of high severity were lower in Period B (OR = 0.41, CI: 0.27-0.61), indicating that the patients were 59% lower in probability of high severity compared to those in Period A.

Based on the random effects estimates results (Table 5), significant variability in baseline severity (intercept variance = 12.32) is observed, indicating the strong individual differences at baseline. The random slope for Visit (variance = 0.40) shows moderate variability, suggesting that patients differ in how their risk of having high disease severity changes over time. Therefore, including both random intercept and slope is reasonable.

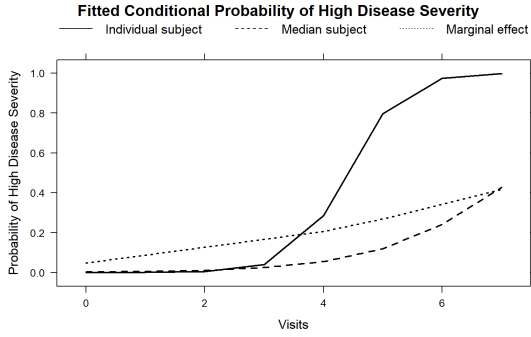


Figure 10: Fitted conditional probability.

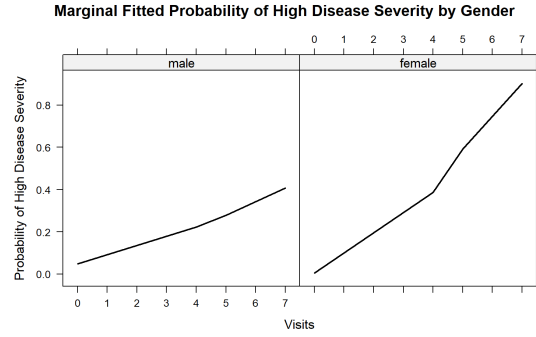


Figure 11: Marginal fitted probability.

Figure 10 and Figure 11 clearly support our conclusion that the risk of high disease severity increases over time. It is worth noting that the effect of gender is pronounced: female patients have a sharp increase in risk from the fourth visit, while the increase for male patients is more gradual.

4 Discussion

In conclusion, the analysis showed that the progression of the disease, measured by both of the Sharp scores (SHS) and disease severity, was influenced by time of Visit, Sex, Period and the interaction of Visit and Sex, which were all statistically significant. Both the mean Sharp scores and probability of high disease severity evolve over time. Although females started with lower baseline levels of joint damage and probability of high severity, they experienced a faster progression over time compared to males. Besides, patients enrolled in the Period B experienced relatively less joint destruction and lower risk of high disease severity.

In addition, both the final models accounted for individual variability, showing substantial differences in patients' baseline joint destruction levels, while the rate of progression over time is relatively consistent across patients.

One limitation of the study is that although collinearity was not formally assessed in this report, potential collinearity issues were observed during the modeling process. This should be addressed in future work, for example by centering the time variable (Visit) or examining variance inflation factors (VIF).

5 Appendix

A Statistical Software

All analyses were conducted using R version 4.3.3. The following R packages were used: `nlme` for fitting linear mixed-effects models using the `lme()` function; `lme4` for generalized linear mixed-effects models using the `glmer()` function; `GLMMadaptive` for flexible GLMM estimation using the `mixed_model()` function; and `ggplot2` for plotting. See the full list of packages and code in the attached R Markdown (.Rmd) file.

B Sharp index Analysis

B1 Initial model with quadratic terms

According to results from Exploratory data analysis (EDA), we chose linear fixed model with polynomial (quadratic) terms to capture non-linear trend and fitted a model with an elaborate fixed-effects. The fitted equation is:

$$\begin{aligned}
 SHS_{ij} = & (\beta_0 + b_{0i}) + \beta_1 \text{Period}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{SNP.f1}_i + \beta_5 \text{SNP.f2}_i \\
 & + \beta_6 \text{Visit}_{ij} + \beta_7 \text{Visit}_{ij}^2 + \beta_8 (\text{Period}_i \times \text{Visit}_{ij}) + \beta_9 (\text{Period}_i \times \text{Visit}_{ij}^2) + \beta_{10} (\text{Age}_i \times \text{Visit}_{ij}) \\
 & + \beta_{11} (\text{Age}_i \times \text{Visit}_{ij}^2) + \beta_{12} (\text{Sex}_i \times \text{Visit}_{ij}) + \beta_{13} (\text{Sex}_i \times \text{Visit}_{ij}^2) + \beta_{14} (\text{SNP.f1}_i \times \text{Visit}_{ij}) \\
 & + \beta_{15} (\text{SNP.f2}_i \times \text{Visit}_{ij}) + \beta_{16} (\text{SNP.f1}_i \times \text{Visit}_{ij}^2) + \beta_{17} (\text{SNP.f2}_i \times \text{Visit}_{ij}^2) + \epsilon_{ij}
 \end{aligned} \tag{3}$$

However, diagnostic checks revealed a systematic trend in the residual plot (Figure 12) and strong skewness in the QQ plot (Figure 13), indicating that the assumptions of normality and homogeneity of variance were not satisfied. One approach to deal with this problem is to transform data. Therefore, we applied a logarithmic transformation on it, $\log(SHS + 1)$, which we denoted as $\log_SHS$.

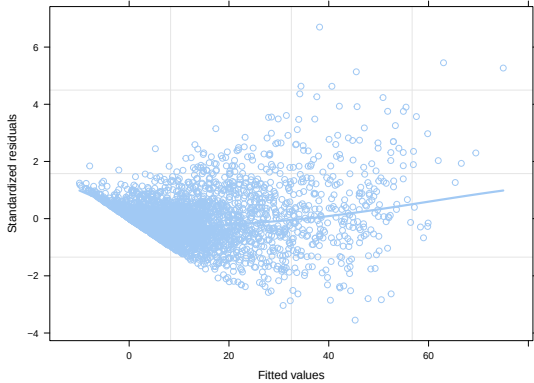


Figure 12: Residuals vs Fitted values_quadratic.

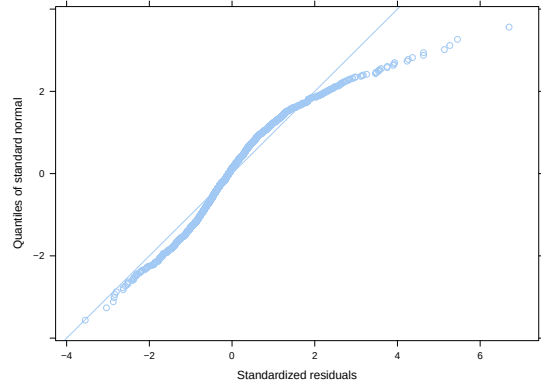


Figure 13: QQ plot_quadratic.

B2 Model Building and Selection

We followed a three-step procedure to build the model.

Full model equation

We constructed an elaborate fixed-effects part that incorporated all relevant covariates and potential interaction terms. Specifically, we modeled $\log_SHS$ as a function of Visit, Period, Age, Sex, and SNP,

including all interactions between visit and the other covariates(named model_slope), the equation is:

$$\begin{aligned} \log _SHS_{ij} = & \beta_0 + \beta_1 \text{Period}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{SNP.f1}_i + \beta_5 \text{SNP.f2}_i + \beta_6 \text{Visit}_{ij} + \\ & \beta_7 (\text{Period}_i \times \text{Visit}_{ij}) + \beta_8 (\text{Age}_i \times \text{Visit}_{ij}) + \beta_9 (\text{Sex}_i \times \text{Visit}_{ij}) + \\ & \beta_{10} (\text{SNP.f1}_i \times \text{Visit}_{ij}) + \beta_{11} (\text{SNP.f2}_i \times \text{Visit}_{ij}) + b_{0i} + b_{1i} \text{Visit}_{ij} + \varepsilon_{ij}, \end{aligned} \quad (4)$$

where $i \in \{1, \dots, 500\}$ (subjects), $j \in \{0, \dots, 7\}$ (visits).

$$\begin{bmatrix} b_{i0} \\ b_{i1} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{\text{int}}^2 & \sigma_{\text{int,visit}} \\ \sigma_{\text{int,visit}} & \sigma_{\text{visit}}^2 \end{bmatrix} \right), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \text{ independent of the covariates.}$$

Hypothesis testing for covariances

we evaluated different random-effects structures by comparing the model with one random effect to the one with two random effects, using an approximate likelihood ratio test (LRT). We fit another elaborate mean model with just random intercept (named model_intercept), the equation is:

$$\begin{aligned} \log _SHS_{ij} = & \beta_0 + \beta_1 \text{Period}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{SNP.f1}_i + \beta_5 \text{SNP.f2}_i + \beta_6 \text{Visit}_{ij} + \\ & \beta_7 (\text{Period}_i \times \text{Visit}_{ij}) + \beta_8 (\text{Age}_i \times \text{Visit}_{ij}) + \beta_9 (\text{Sex}_i \times \text{Visit}_{ij}) + \\ & \beta_{10} (\text{SNP.f1}_i \times \text{Visit}_{ij}) + \beta_{11} (\text{SNP.f2}_i \times \text{Visit}_{ij}) + b_{0i} + \varepsilon_{ij}. \end{aligned} \quad (5)$$

Because the model including two random effects and the model with only a random intercept are nested, they can be compared using the likelihood ratio test (LRT) method, where $\text{LRT} \sim 0.5 \chi_1^2 + 0.5 \chi_2^2$. Our hypothesis test is formulated as follows:

$$H_0 : \sigma_{\text{int,visit}} = \sigma_{\text{visit}}^2 = 0 \quad \text{versus} \quad H_A : \sigma_{\text{visit}}^2 > 0 \text{ or } \sigma_{\text{int,visit}}^2 \neq 0$$

We obtained a p-value of $1.497103e - 05$, which is statistically significant. Therefore, we concluded that the model including the two random effects (model_slope) provides a better fit.

Hypothesis testing for fixed-effects

By checking t_table of the estimated fixed effects parameters of model_slope(Table 7), we could see that the p_value of Age, SNP.f1, SNP.f2, PeriodB:Visit, Age:Visit, SNP.f1:Visit, SNP.f2:Visit are all above 0.05, which means they are not statistically significant, so we moved those terms and fit a simplified model.

Table 7: Fixed Effects Parameter Estimates_model slope

	Estimate	Std. Error	DF	t-value	p-value
(Intercept)	1.8607	0.1565	2217	11.8918	0.0000
PeriodB	-0.2909	0.0621	494	-4.6834	0.0000
Age	-0.0041	0.0023	494	-1.7978	0.0728
sexfemale	-0.5709	0.0619	494	-9.2177	0.0000
SNP.f1	0.0476	0.0712	494	0.6692	0.5037
SNP.f2	0.0464	0.0828	494	0.5602	0.5756
Visit	0.1597	0.0293	2217	5.4436	0.0000
PeriodB:Visit	-0.0097	0.0116	2217	-0.8409	0.4005
Age:Visit	0.0005	0.0004	2217	1.1325	0.2575
sexfemale:Visit	0.2471	0.0115	2217	21.4334	0.0000
SNP.f1:Visit	-0.0106	0.0132	2217	-0.8057	0.4205
SNP.f2:Visit	-0.0241	0.0155	2217	-1.5525	0.1207

Hence we get our final model:

$$\log _SHS_{ij} = \beta_0 + \beta_1 \text{Period}_i + \beta_2 \text{Sex}_i + \beta_3 \text{Visit}_{ij} + \beta_4 (\text{Sex}_i \times \text{Visit}_{ij}) + b_{0i} + b_{1i} \text{Visit}_{ij} + \varepsilon_{ij}$$

B3 Model diagnostics

To verify the assumptions of normality, homoscedasticity and independence, we examined residual plots versus fitted values in overall population (Figure 14), there was no systematic pattern in the residuals, and they were mostly scattered around zero, supporting the assumptions of independence and homoscedasticity. We also inspected the normalized conditional residuals (again in the overall population), which did not reveal any systematic deviations over time or across groups, as shown in Figure 15. In addition, histograms of the random effects (Figure 16 and Figure 17) showed only a few outliers, with most values centered around zero and appearing to be normally distributed. Lastly, we used a QQ plot (Figure 18) to check normality, where the points fell approximately along a straight line, indicating no evidence of skewness or heavy tails.

After fitting the linear mixed-effects model with random intercepts and slopes, we checked whether the model captured the correlation between repeated measurements from the same patient. We inspected the autocorrelation of the residuals to ensure that no additional correlation structure was needed (Figure 19). Although small negative autocorrelations were observed at the first few lags, they remained within the 95% confidence bounds. This indicated no statistically significant residual autocorrelation. Therefore, the current random intercept and slope structure sufficiently captured the within-subject correlation, and no additional residual correlation structure was required.

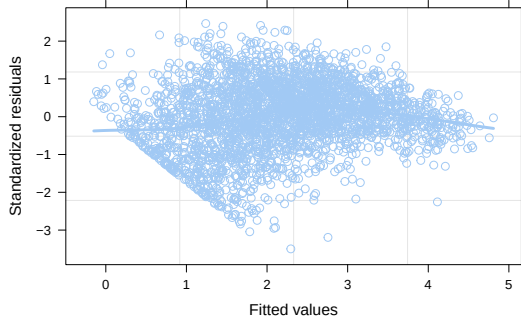


Figure 14: Residuals vs Fitted values.

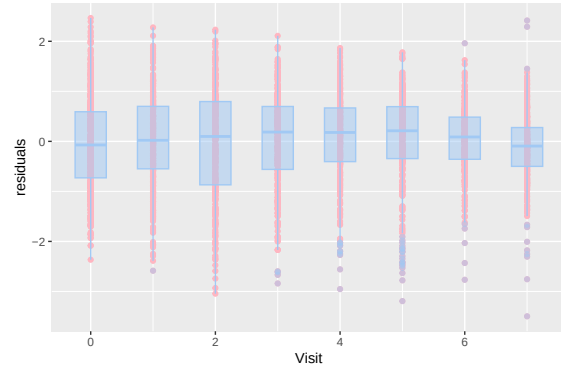


Figure 15: Normalized Conditional Residuals.

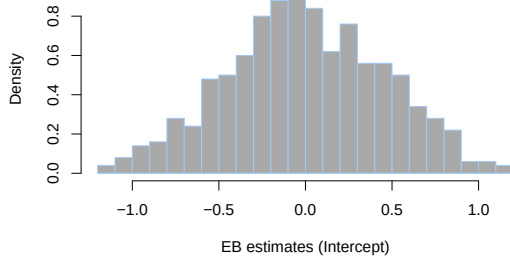


Figure 16: EB Estimates of random Intercepts.

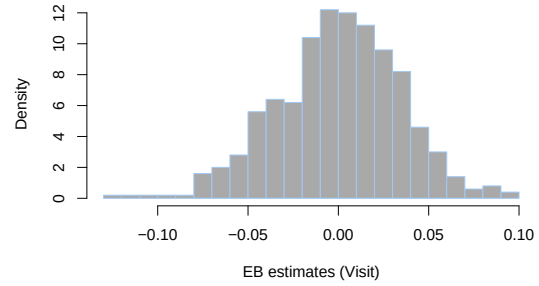


Figure 17: EB Estimates of random Slopes.

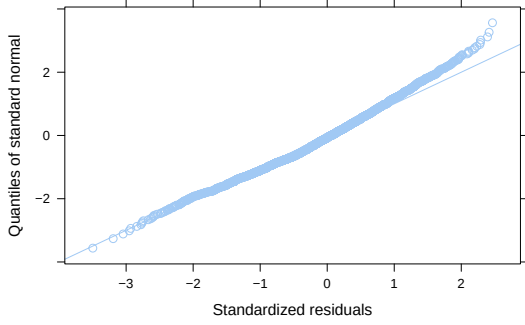


Figure 18: QQ plot.

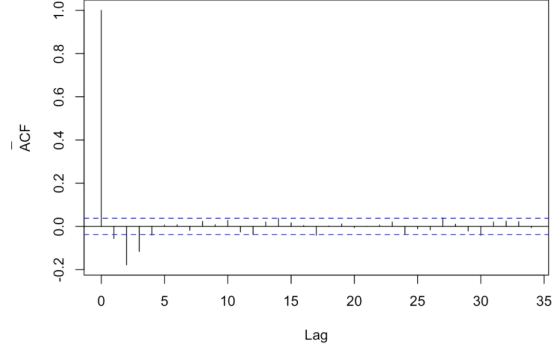


Figure 19: Autocorrelation

C Disease Severity Analysis

C1 Model building Procedures

The modeling procedure started with an elaborated generalized linear mixed model (GLMM) fitted using the `glmer()` function. The model was then simplified step by step to obtain a parsimonious version under model selection criteria. Models were selected based on LRT and AIC. Variables with non-significant effects based on the Wald test were removed one by one.

Note that starting with an extremely elaborate GLMM led to fail to converge. For example, including quadratic terms for both Visit and Age, or a third-degree polynomial for Visit, can cause instability. To improve convergence and interpretability, such terms were excluded.

Model 1 We initially fitted an elaborated model 1 with polynomial term:

$$\begin{aligned} \log \left[\frac{\pi_{ij}}{1 - \pi_{ij}} \right] = & \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Visit}_{ij}^2 + \beta_3 \text{Sex}_i \\ & + \beta_4 (\text{Visit}_{ij} \times \text{Sex}_i) + \beta_5 (\text{Visit}_{ij}^2 \times \text{Sex}_i) \\ & + \beta_6 \text{Age}_i + \beta_7 \text{SNP}_i + \beta_8 \text{Period}_i + b_{0i} + b_{1i} \text{Visit}_{ij} + \epsilon_{ij} \end{aligned} \quad (6)$$

Model 2

We then removed the polynomial term for Visit to evaluate whether a simpler model would be sufficient.

$$\log \left[\frac{\pi_{ij}}{1 - \pi_{ij}} \right] = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Sex}_i + \beta_3 (\text{Visit}_{ij} \times \text{Sex}_i) + \beta_4 \text{Age}_i + \beta_5 \text{SNP}_i + \beta_6 \text{Period}_i + b_{0i} + b_{1i} \text{Visit}_{ij} + \epsilon_{ij} \quad (7)$$

Model 1 and Model 2 were compared using a likelihood ratio test (LRT). Since the p-value is larger than 0.05, the null hypothesis could not be rejected. The more parsimonious Model 2 is therefore preferred.

Table 8: Model Comparison Between Model 1 and Model 2

Model	npar	AIC	BIC	logLik	Deviance	Chisq	Df	Pr(>Chisq)
Model 2	11	1659.0	1724.0	-818.51	1637.0	—	—	—
Model 1	13	1677.6	1754.5	-825.81	1651.6	0	2	1.00

The results of model 2 are summarized in Table 9.

Table 9: Fixed Effects Estimates from the GLMM

Term	Estimate	Std. Error	z value	Pr(> z)	Signif.
(Intercept)	-4.46821	0.76330	-5.854	4.80e-09	***
Visit	0.71755	0.07475	9.599	1.2e-16	***
Sex (female)	-2.26160	0.61090	-3.702	0.000214	***
Age	-0.01181	0.01018	-1.159	0.246296	
SNP.f1	0.18682	0.31540	0.592	0.553619	
SNP.f2	-0.36751	0.37686	-0.975	0.329459	
Period B	-1.39248	0.28193	-4.939	7.85e-07	***
Visit × Sex (female)	1.07287	0.12357	8.682	1.2e-16	***

Model 3

We then removed the variable SNP and Age which are not significant in model 2. Thus model 3 was fitted.

$$\log \left[\frac{\pi_{ij}}{1 - \pi_{ij}} \right] = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Sex}_i + \beta_3 \text{Period}_i + \beta_4 (\text{Visit}_{ij} \times \text{Sex}_i) + b_{0i} + b_{1i} \text{Visit}_{ij} + \epsilon_{ij} \quad (8)$$

Model 2 and Model 3 were compared using a likelihood ratio test (LRT). Since the p-value is larger than 0.05, the null hypothesis could not be rejected. The more parsimonious Model 3 is therefore preferred.

Table 10: Model Comparison Between Model 2 and Model 3

Model	npar	AIC	BIC	logLik	Deviance	Chisq	Pr(> Chisq)
model 3	8	1647.8	1695.1	-815.92	1631.8	—	—
model 2	11	1659.0	1724.0	-818.51	1637.0	0.00	1.00

The results of model 3 are summarized in Table 11.

Table 11: Fixed Effects Estimates from the Final GLMM

	Estimate	Std. Error	z value	Pr(> z)	Signif.
(Intercept)	-6.2552	0.6944	-9.008	$< 2e^{-16}$	***
Visit	0.8516	0.1152	7.391	$1.45e^{-13}$	***
Sex (female)	-3.0193	0.7450	-4.053	$5.06e^{-05}$	***
Period B	-1.5714	0.3474	-4.523	$6.09e^{-06}$	***
Visit × Sex (female)	1.3332	0.1948	6.844	$7.71e^{-12}$	***

Model 4

We then removed the random slope to check whether it was needed. Thus, Model 4 was fitted.

$$\log \left[\frac{\pi_{ij}}{1 - \pi_{ij}} \right] = \beta_0 + \beta_1 \text{Visit}_{ij} + \beta_2 \text{Sex}_i + \beta_3 \text{Period}_i + \beta_4 (\text{Visit}_{ij} \times \text{Sex}_i) + b_{0i} + \epsilon_{ij} \quad (9)$$

Model 3 and Model 4 were compared using a likelihood ratio test (LRT). Since the p-value is smaller than 0.05, the null hypothesis is rejected. Model 3 is therefore preferred, indicating that the random-slope is necessary.

Table 12: Model Comparison Between Model 3 and Model 4

Model	npar	AIC	BIC	logLik	Deviance	Chisq	Pr(> Chisq)
model 4	6	1655.9	1691.4	-821.96	1643.9	—	—
model 3	8	1647.8	1695.1	-815.92	1631.8	12.073	0.002389 **

Since all parameters are statistically significant, Model 3 was selected as the final model.

Once the final model was confirmed, we used **GLMMadaptive** to examine the random effects component of the model since it uses Adaptive Gaussian-Hermite Quadrature (AGHQ) and in return provides a more accurate approximation of integrals over random effects. With the help of GLMMadaptive results we have also cross-checked the stability and reliability of the results obtained from **glmer()**.

Since the primary interest was in the marginal effect of high disease severity in relation to different covariates and their changes over time, the model was further refitted using the **mixed_model()** function. Finally, we used the **marginal_coefs()** function to obtain the coefficient estimates from the marginal model and to calculate the corresponding odds ratios. The results of the marginal model are summarized in Table 6.

C2 Graphical Diagnostics for Random Effects

The random intercepts as well as the random slopes generally follow a normal distribution with acceptable skewness (Figure 21). This supports the normality assumption of random effects in the model. As shown in Figure 20, there is substantial between-subject difference in baseline. Thus including random intercepts is essential in modeling.

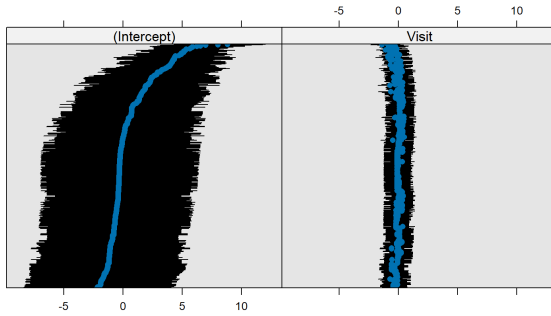


Figure 20: Caterpillar plot for EB estimates of the random effects.

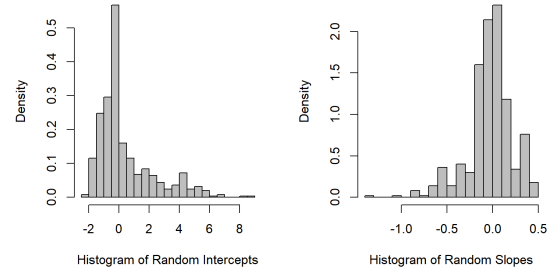


Figure 21: Histograms of EB estimates for random intercepts and slopes.